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Since there is no p.d. across the voltmeter, the p.d.s across 14.4Ω and 4.80Ω resistors are the same.

$$\frac{5.00}{R'} = \frac{2.00}{4.80}$$

$$R' = 12.0\Omega$$

Hence

$$\left(\frac{1}{R} + \frac{1}{14.4} \right)^{-1} = 12.0\Omega$$

$$R = 72.0\Omega$$

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$$B_S = \mu_0 n I = 4\pi \times 10^{-7} \times 15 \times 0.20$$